Question:

Calculate the quantity of heat required to (a) convert a liter of H_2O at 30°C to a liter of water at 60°C, and (b) heat a 1 kg block of aluminum from 30° to 60°C. Assume the specific heat of water and aluminum is, re-spectively, 1 cal/g°C and .215 cal/g°C.

Answer Choices:

A. 28 Kcal, 6.3 Kcal

B. 40 Kcal, 5 Kcal

C. 30 Kcal, 8.58 Kcal

D. 25 Kcal, 8 Kcal

E. 29 Kcal, 7.77 Kcal

F. 34 Kcal, 6.9 Kcal

G. 30 Kcal, 6.45 Kcal

H. 32 Kcal, 7 Kcal

I. 35 Kcal, 7.5 Kcal

J. 31 Kcal, 6.15 Kcal

Reasoning + Answer:

<think>

To calculate the quantity of heat required to convert 1 liter of H2O at 30Â°C to 1 liter of water at 60Â°C, we need to apply the formula Q = mcÎĶT, where m is the mass of the water, c is the specific heat of water (1 cal/gÂ°C), and ÎĶT is the change in temperature.

1 liter of water has a mass of approximately 1000 grams. Thus, Q = 1000g ÃĹ 1 cal/gÂ°C ÃĹ 30Â°C = 30 Kcal.

To calculate the quantity of heat required to heat a 1 kg block of aluminum from 30Â°C to 60Â°C, we apply the same formula Q = mcÎĶT. The specific heat of aluminum is .215 cal/gÂ°C, so Q = 1kg ÃĹ 0.215 cal/gÂ°C ÃĹ 30Â°C = 6.45 Kcal.

</think><answer> G. 30 Kcal, 6.45 Kcal</answer>